The examination is being carried out on the **following application documents**

**Description, Pages**

1-87 filed with entry into the regional phase before the EPO

**Claims, Numbers**

1-15 filed with entry into the regional phase before the EPO

**Drawings, Sheets**

1/8-8/8 filed with entry into the regional phase before the EPO

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure :

| | |
|---|---|
| D1 | TSAI CHING-CHIH ET AL: "Adaptive Motion Control of a Terrain-Adaptive Self-Balancing Leg-Wheeled Mobile Robot over Rough Terrain", 2022 INTERNATIONAL AUTOMATIC CONTROL CONFERENCE (CACS), IEEE, 3 November 2022 (2022-11-03), pages 1-6, XP034242089, DOI: 10.1109/CACS55319.2022.9969857 [retrieved on 2022-12-07] |
| D2 | XU DASHENG ET AL: "End-effector Stabilization of Manipulator Based on Sliding Mode Controller and Disturbance Observer", PROCEEDINGS OF THE 2022 5TH INTERNATIONAL CONFERENCE ON E-BUSINESS, INFORMATION MANAGEMENT AND COMPUTER SCIENCE, ACMPUB27, NEW YORK, NY, USA, 16 December 2022 (2022-12-16), pages 487-493, XP098582130, DOI: 10.1145/3584376.3584464 ISBN: 978-1-4503-9804-6 |
| D3 | ZHOU HAITAO ET AL: "Configuration Transformation of the Wheel-Legged Robot Using Inverse Dynamics Control", 2021 IEEE INTERNATIONAL CONFERENCE ON ROBOTICS |

AND AUTOMATION (ICRA), IEEE, 30 May 2021 (2021-05-30), pages 3091-3096, XP033990478, DOI: 10.1109/ICRA48506.2021.9560781 [retrieved on 2021-10-05]

D4 FUKUSHIMA HIROAKI ET AL: "Sliding-Mode Control for Transformation to an Inverted Pendulum Mode of a Mobile Robot With Wheel-Arms", IEEE TRANSACTIONS ON INDUSTRIAL ELECTRONICS, IEEE SERVICE CENTER, PISCATAWAY, NJ, USA, vol. 62, no. 7, 1 July 2015 (2015-07-01), pages 4257-4266, XP011581615, ISSN: 0278-0046, DOI: 10.1109/TIE.2014.2384475 [retrieved on 2015-05-15]

## 1 Clarity and Conciseness (Art. 84 EPC)

The application does not meet the requirements of Article 84 EPC, because claims 1 and 12 are not clear. Indeed, the feature "sliding surface" is not clear and not defined. Moreover, the feature "the state parameter value of the wheel-legged robot approaching a stable value on the sliding surface" is not clear and not accurate because of the wording "approaching", as it is an objective of the sliding-mode control to keep the parameter value stable on the sliding surface. A solution could be to introduce in claim 1 the feature of the sliding-mode control.

## 2 Novelty and Inventive Step (Art. 52(1), 54 and 56 EPC)

### 2.1 Independent claim 1 - Lack of novelty

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 does not seem novel within the meaning of Article 56 EPC.

D1 discloses (see Chapter III. Backstepping sliding-mode motion control and figures 1, 2 and 6):

- a balance control method for a wheel-legged robot (*see figure 1*), the method being executed by a computer device,

- the wheel-legged robot comprising: a moving wheel, n links, and n rotating joints, the moving wheel being connected to a first link of the n links through a first rotating joint of the n rotating joints, and the n links being connected in series through n-1 rotating joints other than the first rotating joint, where n being a positive integer greater than or equal to 2;
- the method comprising:
- acquiring a state parameter value (*see figure 2, the parameters acquired being inertial measures, angular positions of the joints and wheels*) of the wheel-legged robot, the state parameter value indicating a motion state of the wheel-legged robot at the first time point;
- determining dynamics model parameters (*see chapter III. A. Dynamic Modeling, wherein the "dynamic part is concerned with the interaction between the three accelerations and the two decomposed torque controls"*) for the wheel-legged robot according to a dynamics motion model and the state parameter value, the dynamics model parameters defining a mapping relationship between an angular acceleration at the first time point and a rotation torque at a second time point, the angular acceleration at the first time point comprising angular accelerations of the n links and an angular acceleration of the moving wheel, and the rotation torque at the second time point comprising rotation torques of the n rotating joints;
- establishing a sliding surface according to the state parameter value (*see chapter III. B. Stable trajectory Tracking Control, wherein the sliding surfaces are defined with their equations*), the state parameter value of the wheel-legged robot approaching a stable value on the sliding surface;
- calculating the rotation torques of the n rotating joints according to the sliding surface and the dynamics model parameters (*see equations (4) and (5) and figure 2*) ; and
- controlling the n rotating joints according to the rotation torques of the n rotating joints at the second time point (*see figure 2 and chapter III. B. Stable trajectory Tracking Control*).

All the features of claim 1 are thus disclosed by D1.

## 2.2 Independent claims 12-15 - Lack of novelty

The present application does not meet the requirements of Article 52(1) EPC

because the claims 12-15 are directly and unambiguously derivable from method claim 1, thus the reasoning applied for claim 1 can be applied here mutatis mutandis.

## 2.3 Dependent claims 4, 5, 8, 11 - Lack of novelty or inventive step

The additional features defined in the dependent claims 4, 5, 8, 11 are disclosed in at least one of the documents D1 to D2, or appear to be obvious for the skilled person, and therefore do not seem to impart novelty (claims 5 and 8) or inventive step (claims 4 and 11) to the claimed subject-matter.
In particular :

- claim 4 - inverted pendulum model, Euler-Lagrange equation : see D1 chapter IV.A. for the inverted pendulum model, the use of the Euler-Lagrange equation being obvious for the skilled person.
- claim 5 - n sliding surfaces : see D1 chapter III.B. ; see also D2 wherein a sliding surface is calculated for each joint.
- claim 8 - controlling a rotating motor : see D1, wherein there is a motor for the wheels and a motor for the knees.
- claim 11 - deflection angles via IMU for each link, angular velocities of links and wheel via motor encoders : D1 discloses the use of IMU and the measure of the angular positions, it would seem obvious for the skilled person to use a IMU for each link and to use motor encoders for the angular positions of each link and wheel and derivate it for getting the velocity.

## 3 Other deficiencies

To meet the requirements of Rule 42(1)(b) EPC, please cite D1 to D4 in the description and indicate the relevant background art disclosed in them.

## 4 Procedure

4.1 Please file new claims which take account of the above comments.

4.2 In order to comply with the requirements of Rule 137(4) EPC, please clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see EPC Guidelines H-III, 2.1).

4.3 Amendments should be made by filing replacement pages. Unnecessary recasting of the description should be avoided. An amended abstract is not required. Attention is drawn to the requirements of Rule 50(1) EPC. According to Rule 50(1) in conjunction with Art. 2(7) of the decision of the President of the EPO dated 5 July 2025 (OJ EPO 2025, A49), amendments must be typed or printed.